

6. Control Chart Constants - Six Sigma Demystified, Second Edition, 2nd Edition [Book]

appendix 6Control Chart Constants

CHART FOR AVERAGE				CHART FOR STANDARD DEVIATIONS					
Observations in Sample, n	Factors for Control Limits			Factors for Central Line		Factors for Control Limits			
	A	A_2	A_3	c_4	$1/c_4$	B_3	B_4	B_5	B_6
2	2.121	1.880	2.659	0.7979	1.2533	0	3.267	0	3.267
0	2.606								
3	1.732	1.023	1.954	0.8862	1.1284	0	2.568	0	2.276
4	1.500	0.729	1.628	0.9213	1.0854	0	2.266	0	2.088
5	1.342	0.577	1.427	0.9400	1.0638	0	2.089	0	1.964
6	1.225	0.483	1.287	0.9515	1.0510	0.030	1.970	0.029	1.874
7	1.134	0.419	1.182	0.9594	1.0423	0.118	1.882	0.113	1.806
8	1.061	0.373	1.099	0.9650	1.0363	0.185	1.815	0.179	1.751
9	1.000	0.337	1.032	0.9693	1.0317	0.239	1.761	0.232	1.707
10	0.949	0.308	0.975	0.9727	1.0281	0.284	1.716	0.276	1.669
11	0.905	0.285	0.927	0.9754	1.0252	0.321	1.679	0.313	1.637
12	0.866	0.266	0.886	0.9776	1.0229	0.354	1.646	0.346	1.610
13	0.832	0.249	0.850	0.9794	1.0210	0.382	1.618	0.374	1.585
14	0.802	0.235	0.817	0.9810	1.0194	0.406	1.594	0.399	1.563
15	0.775	0.223	0.789	0.9823	1.0180	0.428	1.572	0.421	1.544
16	0.750	0.212	0.763	0.9835	1.0168	0.448	1.552	0.440	1.526
17	0.728	0.203	0.739	0.9845	1.0157	0.466	1.534	0.458	1.511
18	0.707	0.194	0.718	0.9854	1.0148	0.482	1.518	0.475	1.496
19	0.688	0.187	0.698	0.9862	1.0140	0.497	1.503	0.490	1.483
20	0.671	0.180	0.680	0.9869	1.0133	0.510	1.490	0.504	1.470
21	0.655	0.173	0.663	0.9876	1.0126	0.523	1.477	0.516	1.459
22	0.640	0.167	0.647	0.9882	1.0119	0.534	1.466	0.528	1.448
23	0.626	0.162	0.633	0.9887	1.0114	0.545	1.455	0.539	1.438
24	0.612	0.157	0.619	0.9892	1.0109	0.555	1.445	0.549	1.429
25	0.600	0.153	0.606	0.9896	1.0105	0.565	1.435	0.559	1.420

CHART FOR RANGES
X CHARTS

Observations in Sample, n	Factors for Central Line			Factors for Control Limits				
	d_2	$1/d_2$	d_3	D_1	D_2	D_3	D_4	E_2
2	1.128	0.8865	0.853	0	3.686	0	3.267	2.660
3	1.693	0.5907	0.888	0	4.358	0	2.574	1.772
4	2.059	0.4857	0.880	0	4.698	0	2.282	1.457
5	2.326	0.4299	0.864	0	4.918	0	2.114	1.290
6	2.534	0.3946	0.848	0	5.078	0	2.004	1.184
7	2.704	0.3698	0.833	0.204	5.204	0.076	1.924	1.109
8	2.847	0.3512	0.820	0.388	5.306	0.136	1.864	1.054
9	2.970	0.3367	0.808	0.547	5.393	0.184	1.816	1.010
10	3.078	0.3249	0.797	0.687	5.469	0.223	1.777	0.975
11	3.173	0.3152	0.787	0.811	5.535	0.256	1.744	0.945
12	3.258	0.3069	0.778	0.922	5.594	0.283	1.717	0.921
13	3.336	0.2998	0.770	1.025	5.647	0.307	1.693	0.899
14	3.407	0.2935	0.763	1.118	5.696	0.328	1.672	0.881
15	3.472	0.2880	0.756	1.203	5.741	0.347	1.653	0.864
16	3.532	0.2831	0.750	1.282	5.782	0.363	1.637	0.849
17	3.588	0.2787	0.744	1.356	5.820	0.378	1.622	0.836
18	3.640	0.2747	0.739	1.424	5.856	0.391	1.608	0.824
19	3.689	0.2711	0.734	1.487	5.891	0.403	1.597	0.813
20	3.735	0.2677	0.729	1.549	5.921	0.415	1.585	0.803
21	3.778	0.2647	0.724	1.605	5.951	0.425	1.575	0.794
22	3.819	0.2618	0.720	1.659	5.979	0.434	1.566	0.786
23	3.858	0.2592	0.716	1.710	6.006	0.443	1.557	0.778
24	3.895	0.2567	0.712	1.759	6.031	0.451	1.548	0.770
25	3.931	0.2544	0.708	1.806	6.056	0.459	1.541	0.763